

Project 1: Dockerizing Jenkins Pipeline- Demonstrate continuous integration and delivery by Dockerizing Jenkins Pipeline.

Git Setup & Basic Workflow

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Archisha>cd Documents
C:\Users\Archisha\Documents>cd semVII
C:\Users\Archisha\Documents\semVII>git clone https://github.com/adroitathena2/my-git-practice.git
Cloning into 'my-git-practice'...
warning: You appear to have cloned an empty repository.
C:\Users\Archisha\Documents\semVII>cd my-git-practice
C:\Users\Archisha\Documents\semVII\my-git-practice>echo # My Git Practice Project >
The syntax of the command is incorrect.
C:\Users\Archisha\Documents\semVII\my-git-practice>echo # My Git Practice Project > README.md
C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\Archisha\Documents\semVII\my-git-practice>
```

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "Add initial README.md file"
On branch main

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\Archisha\Documents\semVII\my-git-practice>
```

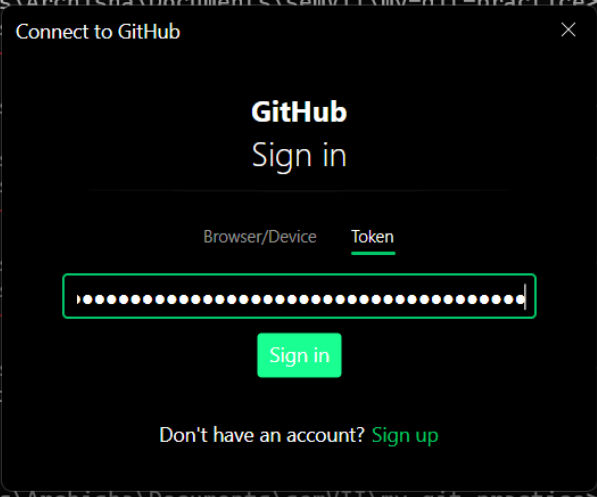
```
C:\Users\Archisha\Documents\semVII\my-git-practice>git add README.md
```

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "Add initial README.md file"
[main (root-commit) 5732f09] Add initial README.md file
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

The screenshot shows the GitHub 'New personal access token (classic)' page. The left sidebar contains the navigation menu with 'Personal access tokens' selected. The main content area has a title 'New personal access token (classic)' and a description: 'Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).' Below this is a 'Note' box with the text 'Windows Command Prompt Git'. The 'Expiration' section shows a dropdown menu set to '30 days (Aug 01, 2026)'. The 'Select scopes' section lists various scopes with checkboxes: 'repo' (checked), 'repo:status', 'repo:deployment', 'public_repo', 'repo:invite', 'security_events', 'workflow', 'write:packages', 'read:packages', 'delete:packages', 'admin:org', 'write:org', 'read:org', and 'manage_runners:org'. Each scope has a brief description of its permissions.

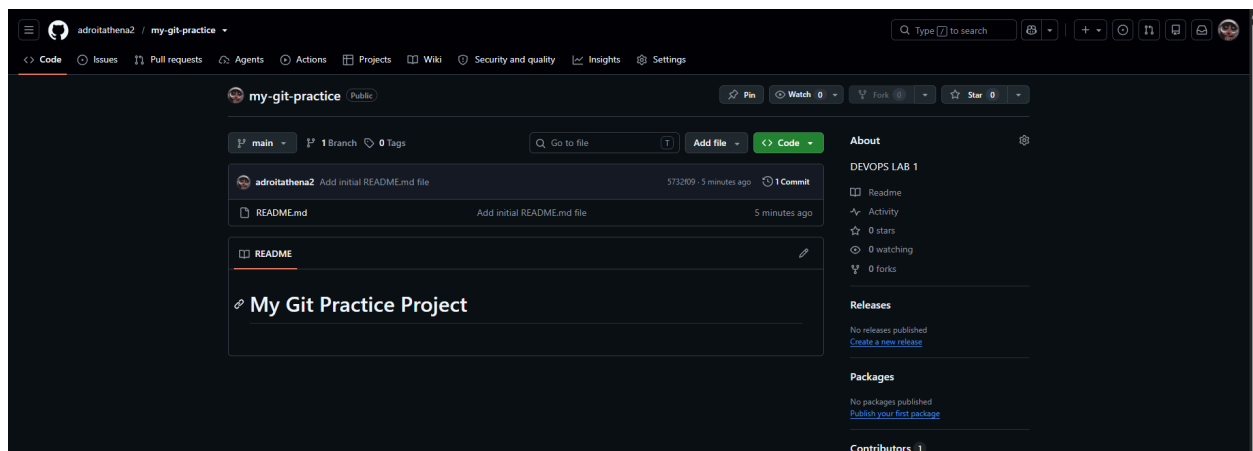
The screenshot shows the GitHub 'Personal access tokens (classic)' page after a token has been generated. A message at the top states: 'Some of the scopes you've selected are included in other scopes. Only the minimum set of necessary scopes has been saved.' The main content area has a title 'Personal access tokens (classic)' and a 'Generate new token' button. Below the title, it says 'Tokens you have generated that can be used to access the [GitHub API](#)'. A notification box says 'Make sure to copy your personal access token now. You won't be able to see it again!'. Below this, a green box displays the token 'ghp_qu54pdq7UkIoo52X9vurKY7qawP0c1207kFa' with a 'Copied' button and a 'Delete' button. A note at the bottom explains that personal access tokens (classic) function like ordinary OAuth access tokens and can be used to authenticate to the API over Basic Authentication.

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git push origin main
error: Connect to GitHub
error: com/adroitathena2/my-git-practice.git'
git add README.md
git push origin main
error: com/adroitathena2/my-git-practice.git'
git push -u origin main
error: com/adroitathena2/my-git-practice.git'
git commit -m "Add initial README.md file"
fatal: Authentication failed for 'https://github.com/adroitathena2/my-git-practice.git/'
C:\Users\Archisha\Documents\semVII\my-git-practice>git push -u origin main
```

A GitHub Sign in dialog box is overlaid on the terminal. It has a title bar "Connect to GitHub" with a close button. The main heading is "GitHub Sign in". Below it are two tabs: "Browser/Device" and "Token", with "Token" selected. There is a password input field with a green border and a "Sign in" button. At the bottom, it says "Don't have an account? Sign up".

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 257 bytes | 257.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/adroitathena2/my-git-practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\Archisha\Documents\semVII\my-git-practice>
```



Git Branching & Merging

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch main
Your branch is up to date with 'origin/main'.
```

```
nothing to commit, working tree clean
```

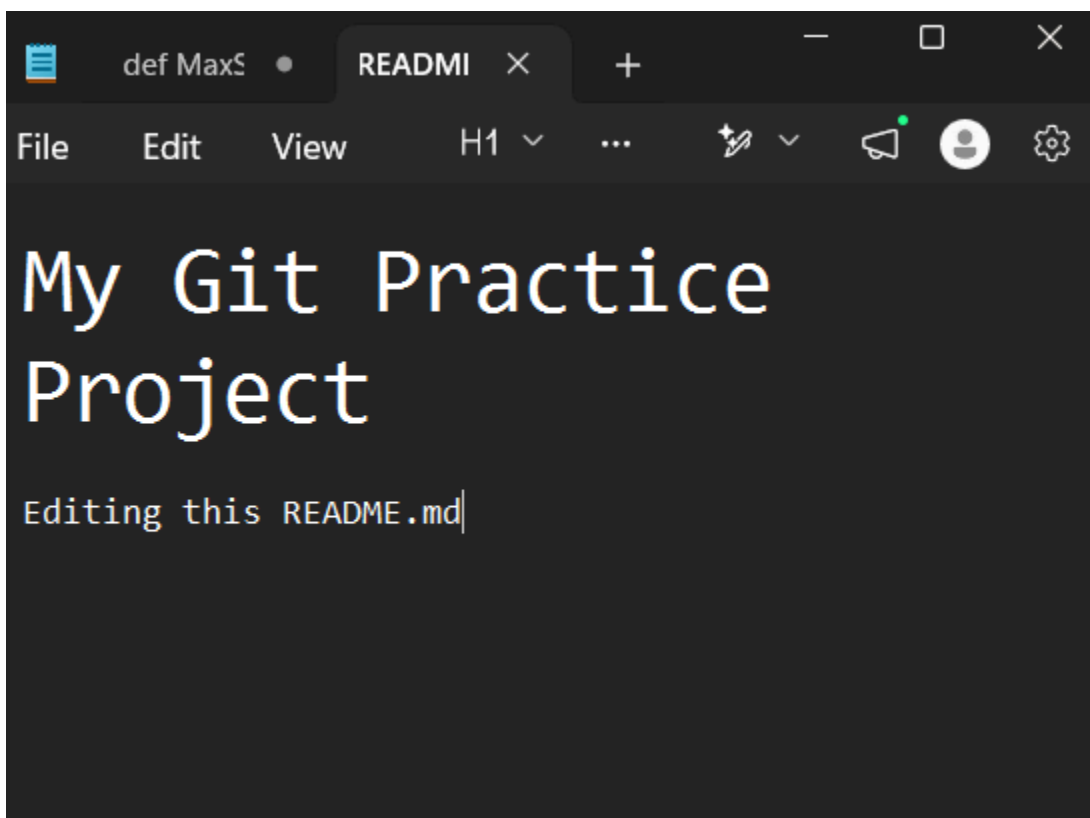
```
C:\Users\Archisha\Documents\semVII\my-git-practice>
```

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git branch feature-update
```

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git checkout feature-update
Switched to branch 'feature-update'
```

```
C:\Users\Archisha\Documents\semVII\my-git-practice>git branch
```

```
* feature-update
main
```



```

C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Archisha\Documents\semVII\my-git-practice>git add README.md

C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "UPDATES README on feature branch"
[feature-update 62f165b] UPDATES README on feature branch
 1 file changed, 6 insertions(+), 1 deletion(-)

C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "UPDATES README on feature branch"
On branch feature-update
nothing to commit, working tree clean

C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch feature-update
nothing to commit, working tree clean

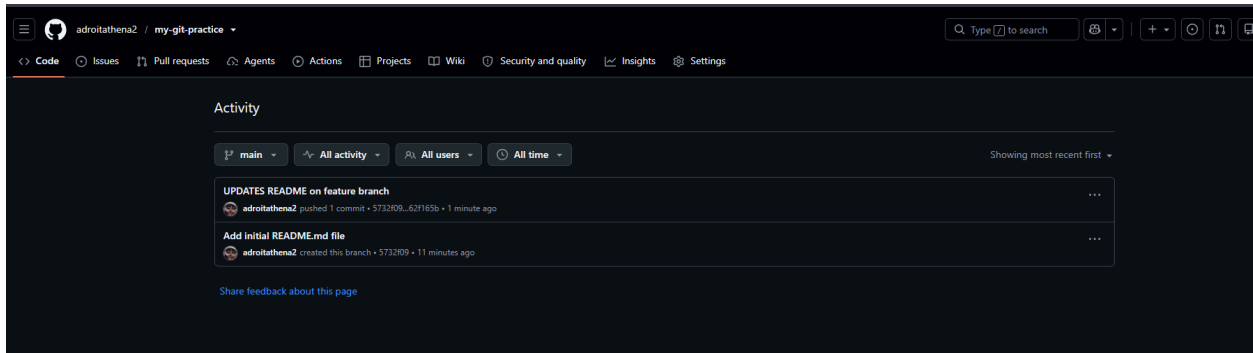
C:\Users\Archisha\Documents\semVII\my-git-practice>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\Archisha\Documents\semVII\my-git-practice>git merge feature-update
Updating 5732f09..62f165b
Fast-forward
 README.md | 7 ++++++-
 1 file changed, 6 insertions(+), 1 deletion(-)

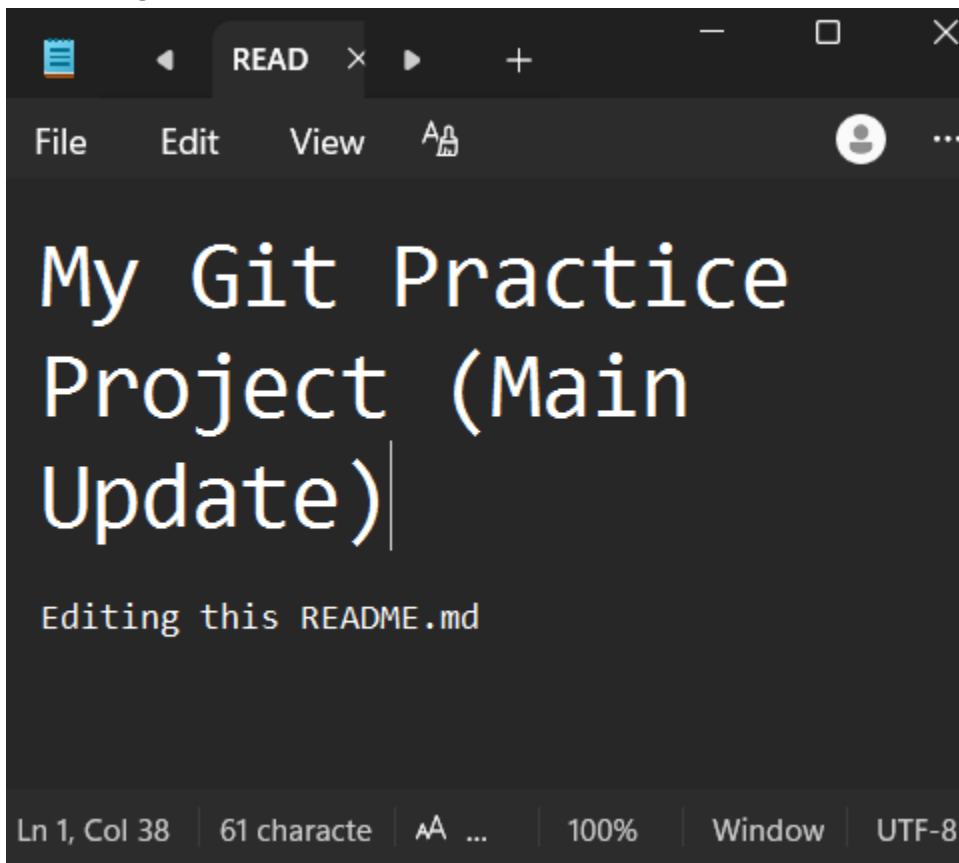
C:\Users\Archisha\Documents\semVII\my-git-practice>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 320 bytes | 320.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/adroitathena2/my-git-practice.git
 5732f09..62f165b  main -> main

```

The screenshot shows the GitHub web interface for a repository named 'my-git-practice' owned by 'adroitathena2'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main' with 1 branch and 0 tags. The repository contains a single file named 'README.md'. The commit history shows two commits: one by 'adroitathena2' titled 'UPDATES README on feature branch' (commit 62f165b, 1 minute ago) and another commit (5732f09). The README file is currently being edited, and the title 'My Git Practice Project' is visible. The right sidebar shows the 'About' section with 'DEVOPS LAB 1' and the 'Releases' and 'Packages' sections, both indicating no published content.



Simulating Git Collaboration & Conflict Resolution



```
C:\Users\Archisha\Documents\semVII\my-git-practice>git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

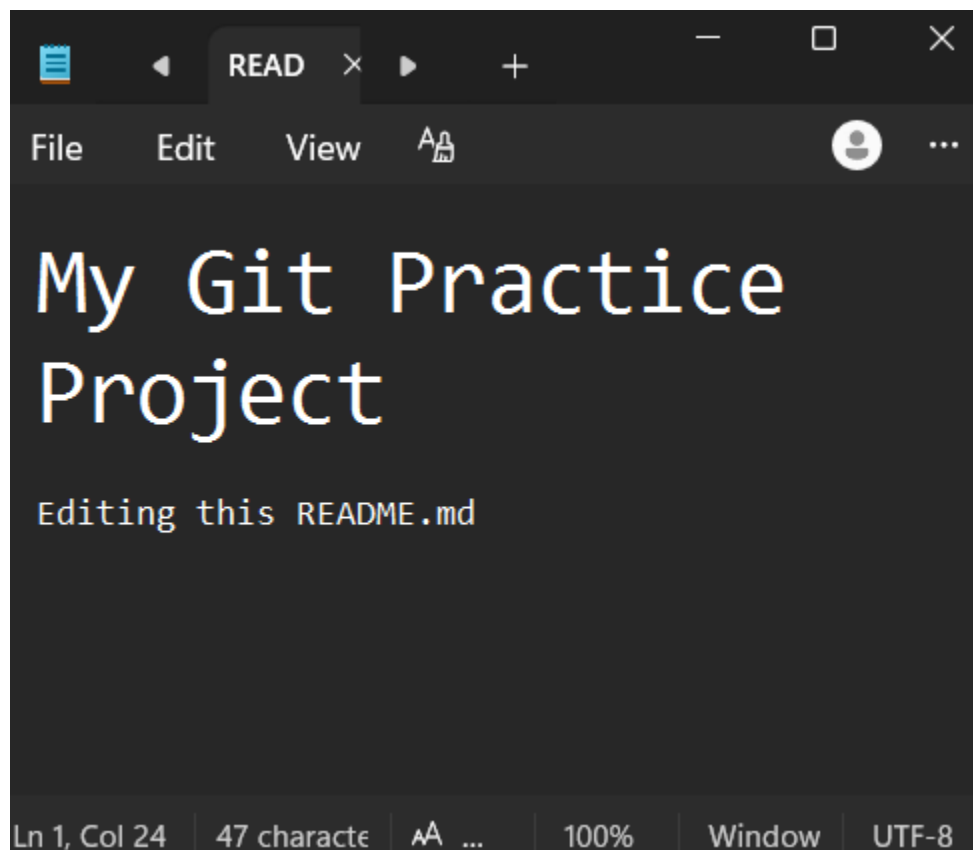
C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch main
Your branch is up to date with 'origin/main'.

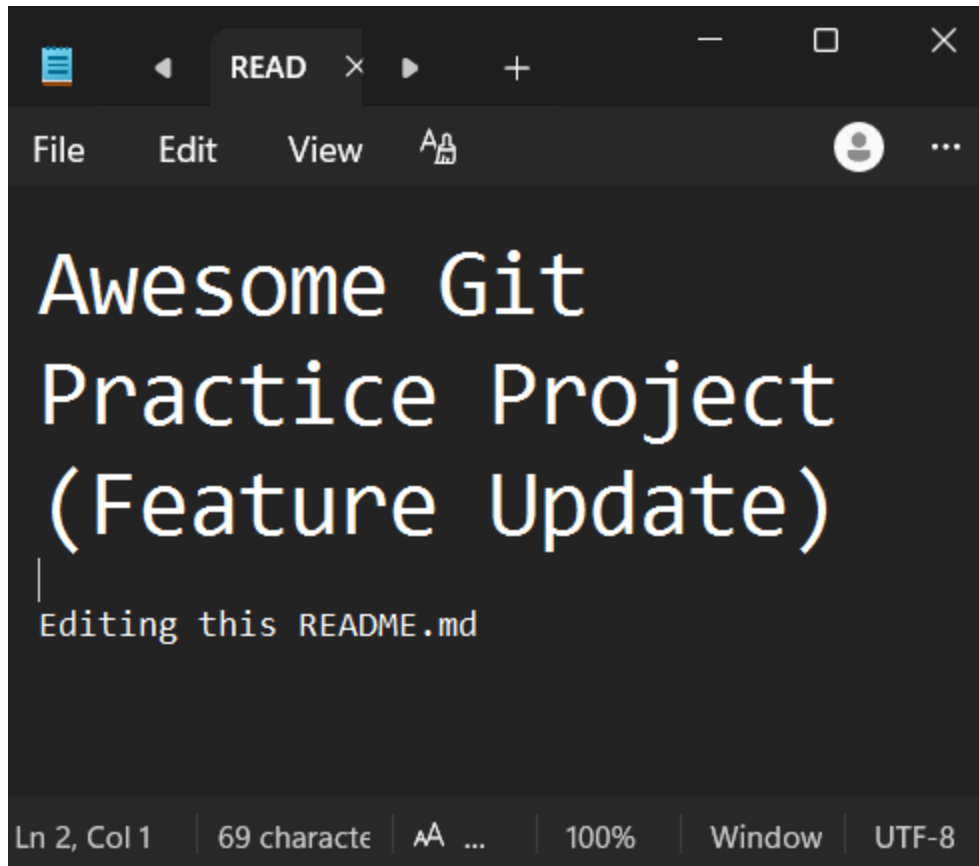
nothing to commit, working tree clean

C:\Users\Archisha\Documents\semVII\my-git-practice>git add README.md

C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "UPDATE README first line on main"
[main 0a582fc] UPDATE README first line on main
1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\Archisha\Documents\semVII\my-git-practice>git checkout -b conflicting-feature
Switched to a new branch 'conflicting-feature'
```





```
C:\Users\Archisha\Documents\semVII\my-git-practice>git add README.md

C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "Update README first line on feature branch"
[conflicting-feature 6786d92] Update README first line on feature branch
1 file changed, 1 insertion(+), 1 deletion(-)

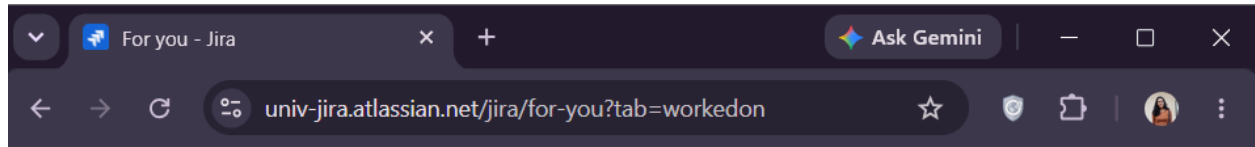
C:\Users\Archisha\Documents\semVII\my-git-practice>git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

C:\Users\Archisha\Documents\semVII\my-git-practice>git merge conflicting-feature
Updating 0a582fc..6786d92
Fast-forward
 README.md | 2 + -
1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\Archisha\Documents\semVII\my-git-practice>git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
```


2. Jira Project & Issue Tracking



[← Back to templates](#)

Name your space

Required fields are marked with an asterisk *

Name *

How your space is managed *

 Team-managed

[Learn more about space types](#)

Space permissions

Choose who can view, edit, and comment

With space permissions, control who has access to your spaces and what they can do.

[Try it free for 14 days](#)

Key * 

Template

[See details](#)

Step 1 of 3

[Next](#)

For you

Recent

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Apps

Explore more apps

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Spaces

Recent

DevOps Lab

Farm-Layout

More spaces

Recommended

Collect requests

Teams

More

Jira

Search

Create

Feedback

Free

Jira / Marketplace apps / GitHub for Atlassian

GitHub for Atlassian

by Atlassian

OVERALL RATINGS

3.3/5

★★★★★

(509)

INSTALLS

142,533

TRUST SIGNALS

CLOUD FORTIFIED

Overview

Reviews

Privacy & Security

Support

development workflows

Hub, Jira, and your

apps having unified data

Atlassian-powered app

Mobile development

6

96min

2d 1h

132

Work items

5 open

3 critical

2 ready for review

Adding GitHub for Atlassian...

We'll tell you when your app has been added.

Close

Unlock

Rovo can work get

Try Rovo now

github.com/organizations/plan

Choose a plan

Pick a plan for your organization

Free

The basics for individuals and organizations

\$0USD per user/month

Create a free organization

Unlimited public/private repositories

Dependabot security and version updates

2,000 CI/CD minutes/month

500MB of Packages storage

Issues & Projects

Community support

FEATURED ADD-ONS

GitHub Codespaces Access

MOST POPULAR

Team

Advanced collaboration for individuals and organizations

\$4USD per user/month

Continue with Team

Everything included in Free, plus...

Access to GitHub Codespaces

Repository rules

Multiple reviewers in pull requests

Draft pull requests

Code owners

Required reviewers

Pages and Wikis

Environment deployment branches and secrets

3,000 CI/CD minutes/month

2GB of Packages storage

Enterprise

Security, compliance, and flexible deployment

Starting at

\$21USD per user/month

Start a free trial

Contact Sales

Everything included in Team, plus...

Data residency

Enterprise Managed Users

User provisioning through SCIM

Enterprise Account to centrally manage multiple organizations

Environment protection rules

Repository rules

Audit Log API

SOC1, SOC2, type 2 reports annually

FedRAMP Tailored Authority to Operate (ATO)

SAML single sign-on

Tell us about your organization

Set up your organization

Organization name *

SIT-DevOps

This will be the name of your account on GitHub.
Your URL will be: <https://github.com/SIT-DevOps>.

Contact email *

archisha.yadav.btech2023@sitpune.edu.in

This organization belongs to:

- ☒ **My personal account**
I.e., adroitathena2 (Archisha Yadav)
- ☐ **A business or institution**
For example: GitHub, Inc., Example Institute, American Red Cross

☒ I hereby accept the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#).

Next



Type to search



Install Atlassian

Install on your organization SIT-DevOps

for these repositories:

☒ **All repositories**
This applies to all current and future repositories owned by the resource owner. Also includes public repositories (read-only).

☐ **Only select repositories**
Select at least one repository. Also includes public repositories (read-only).

with these permissions:

☒ Read access to Dependabot alerts, actions, administration, members, metadata, organization administration, secret scanning alerts, and security events

☒ Read and write access to code, deployments, issues, and pull requests

Install

Cancel

Next: you'll be directed to the GitHub App's site to complete setup.



SIT-DevOps is now connected!

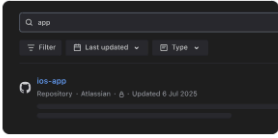
Your GitHub data will now be available to access in Atlassian products.

Connected organizations[Add another organization](#) | [View All](#)

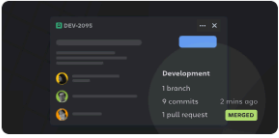
 **SIT-DevOps**
github.com/SIT-DevOps

All repos connected 

Explore GitHub for Atlassian capabilities



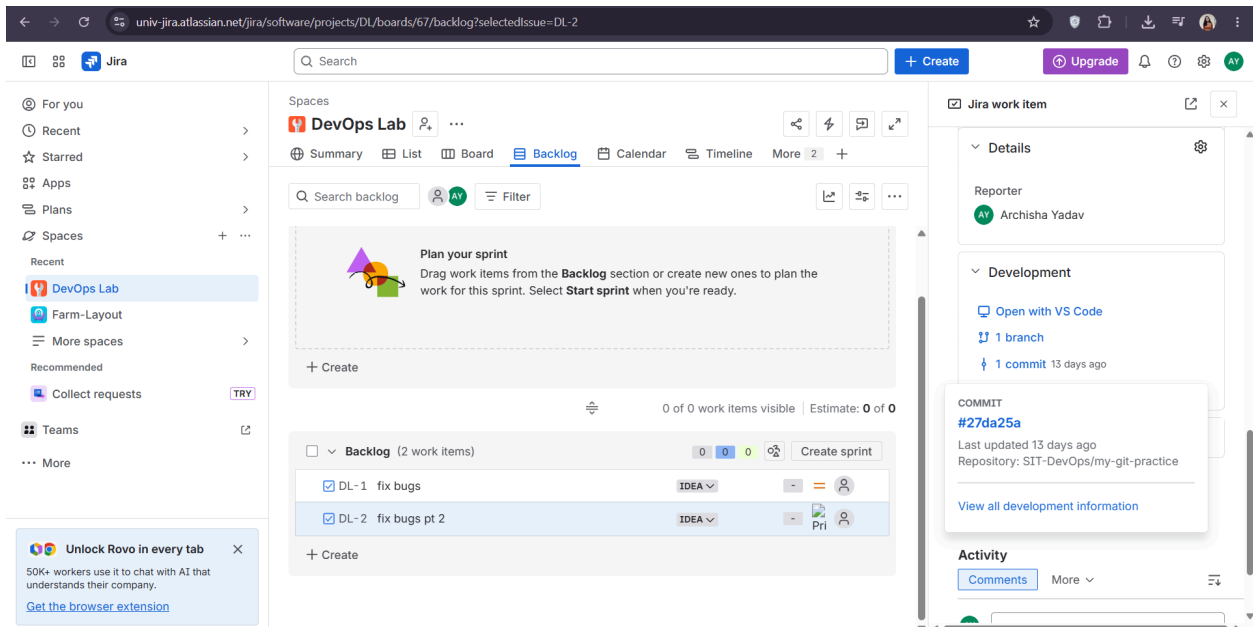
Search for GitHub development information using Rovo Search



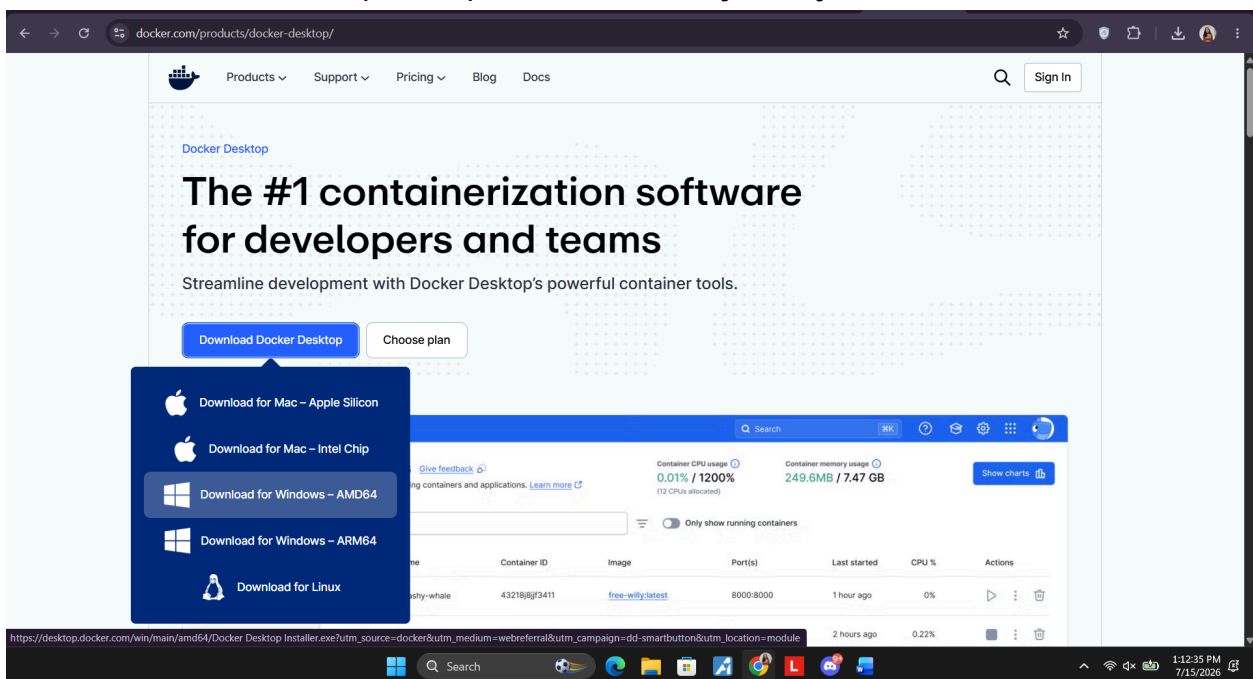
View GitHub development information on Jira work items

[Exit set up](#)

```
C:\Users\Archisha\Documents\semVII>cd my-git-practice
C:\Users\Archisha\Documents\semVII\my-git-practice>git checkout -b DL-2-practice-task
Switched to a new branch 'DL-2-practice-task'
C:\Users\Archisha\Documents\semVII\my-git-practice>git add .
C:\Users\Archisha\Documents\semVII\my-git-practice>git commit -m "DL-2 added a new line to the README file"
On branch DL-2-practice-task
nothing to commit, working tree clean
C:\Users\Archisha\Documents\semVII\my-git-practice>git push -u origin DL-2-practice-task
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (9/9), 1.42 KiB | 1.42 MiB/s, done.
Total 9 (delta 1), reused 9 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'DL-2-practice-task' on GitHub by visiting:
remote:   https://github.com/SIT-DevOps/my-git-practice/pull/new/DL-2-practice-task
remote:
To https://github.com/SIT-DevOps/my-git-practice.git
 * [new branch]      DL-2-practice-task -> DL-2-practice-task
branch 'DL-2-practice-task' set up to track 'origin/DL-2-practice-task'.
C:\Users\Archisha\Documents\semVII\my-git-practice>
```



3. Basic Containerization (Docker) & Jenkins Freestyle Project



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Archisha>docker --version
Docker version 29.6.1, build 8900f1d

C:\Users\Archisha>
```

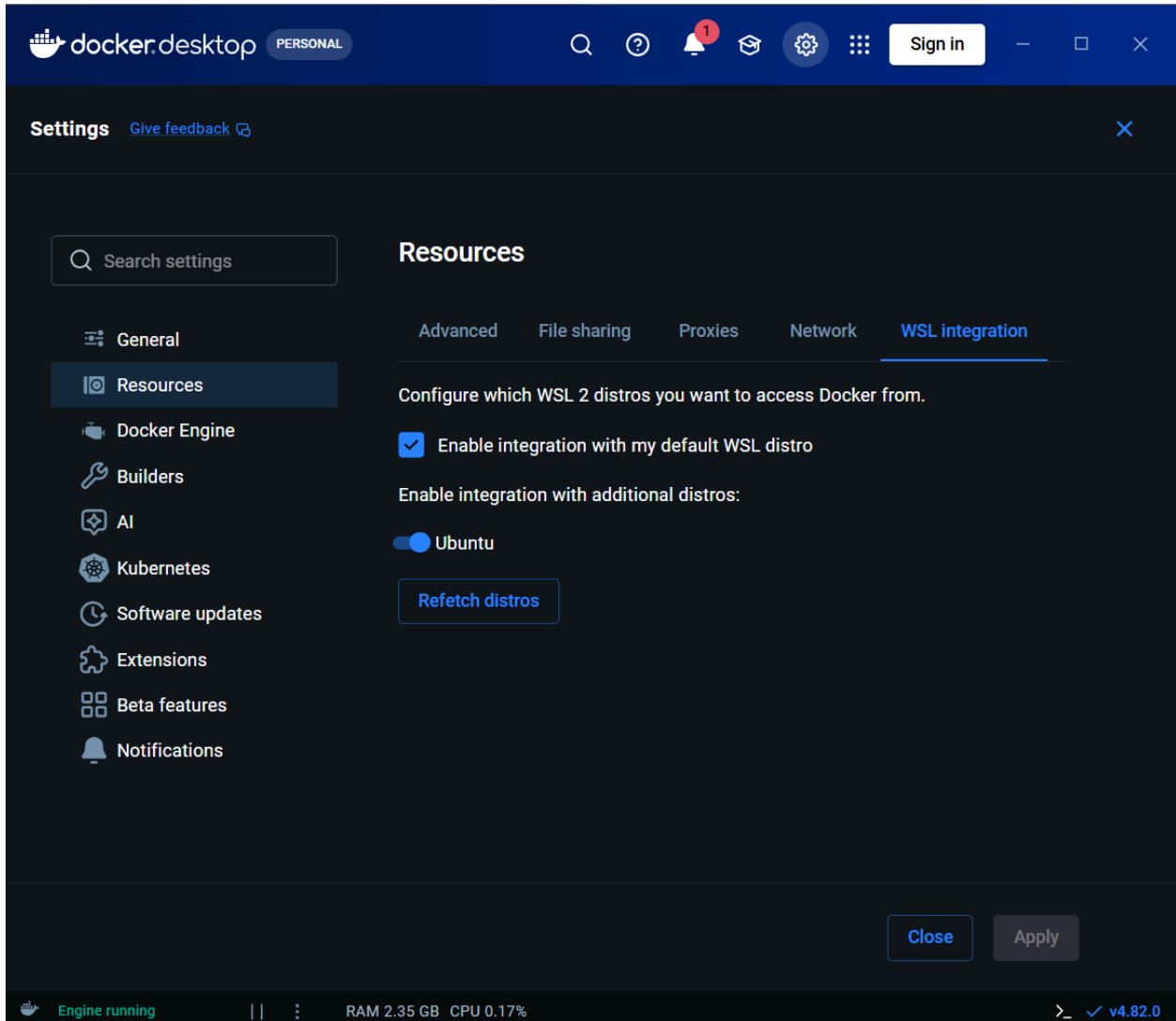
```
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> wsl --install
The requested operation requires elevation.
Downloading: Windows Subsystem for Linux 2.7.10
Installing: Windows Subsystem for Linux 2.7.10
Windows Subsystem for Linux 2.7.10 has been installed.
The operation completed successfully.
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
^[I^[OProvisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: archisha
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Help improve Ubuntu!

Help us improve Ubuntu features and compatibility by sharing system reports with Canonical.
Reports are sent anonymously and do not contain any personal data.
For legal details, please visit: https://ubuntu.com/legal/systems-information-notice

We will save your answer to Windows and will only ask you once.

Would you like to opt-in to platform metrics collection (Y/n)? To see an example of the data collected, enter 'e'.
[Y/n/e]: n
archisha@LAPTOP-FGRFQ556:/mnt/c/WINDOWS/system32$ |
```



```
PS C:\WINDOWS\system32> docker pull jenkins/jenkins:lts-jdk17
lts-jdk17: Pulling from jenkins/jenkins
4d2125ccf71c: Pull complete
6327a7366a6f: Pull complete
46f93bf1cb2f: Pull complete
53c88f1dfeb7: Pull complete
76eac8d3a597: Pull complete
01938f7c040b: Pull complete
6da7907f70f4: Pull complete
0b9e958129f4: Pull complete
503dee4523cf: Pull complete
69503609dc01: Pull complete
f44eb945cec1: Pull complete
b7b154fd69a1: Pull complete
c47cc6af1e21: Download complete
Digest: sha256:cce1eeb79902722f3cf7a8831c41e95c39f4a42b9a2981959630cc66539b3c2d
Status: Downloaded newer image for jenkins/jenkins:lts-jdk17
docker.io/jenkins/jenkins:lts-jdk17

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview jenkins/jenkins:lts-jdk17
PS C:\WINDOWS\system32>

PS C:\WINDOWS\system32> docker run -d -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home --name myjenkins jenkins/jenkins:lts-jdk17
ae7f26357f4a8500b41c4c35bbab3b2585186d7b4e60254f118a799e91153b28
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> docker logs myjenkins
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-07-15 08:42:03.563+0000 [id=1] INFO winstone.Logger$LogInternal: Beginning extraction from war file
2026-07-15 08:42:02.361+0000 [id=1] WARNING o.e.j.eep.nested.ContextHandler$setContextPath: Empty contextPath
2026-07-15 08:42:02.484+0000 [id=1] INFO org.eclipse.jetty.server.Server$doStart: Jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4086d89d7b4881e792d7b7194dcf7b6de4be25d6; jvm 17.0.10-b0
2026-07-15 08:42:02.778+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor$visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-07-15 08:42:02.822+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager$doStart: Session workerName=node0
2026-07-15 08:42:03.126+0000 [id=1] INFO hudson.WebAppMain$contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-07-15 08:42:03.213+0000 [id=1] INFO o.e.j.s.handler.ContextHandler$doStart: Started oeje9n.ContextHandler$CoreContextHandler$66577ba8(Jenkins v2.541.3,,bFile:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$Co
reContextHandler$CoreContextHandler$108adac8(STARTED))
2026-07-15 08:42:03.229+0000 [id=1] INFO o.e.j.s.server.AbstractConnector$doStart: Started oejs.ServerConnector$4e4b6f47(HTTP/1.1, ([http/1.1]){0.0.0.0:8080})
2026-07-15 08:42:03.229+0000 [id=1] INFO org.eclipse.jetty.server.Server$doStart: Started oejs.Server$4aadafe1(STARTING)[12.1.5,sto=0] [825ms
2026-07-15 08:42:03.223+0000 [id=26] INFO winstone.Logger$LogInternal: Winstone Servlet Engine running: controlPort=disabled
2026-07-15 08:42:03.355+0000 [id=25] INFO jenkins.model.Jenkins$init: Starting version 2.541.3
2026-07-15 08:42:03.451+0000 [id=34] INFO jenkins.InitReactorRunner$1$onAttained: Started initialization
2026-07-15 08:42:03.463+0000 [id=34] INFO jenkins.InitReactorRunner$1$onAttained: Listed all plugins
2026-07-15 08:42:04.108+0000 [id=36] INFO jenkins.InitReactorRunner$1$onAttained: Prepared all plugins
2026-07-15 08:42:04.112+0000 [id=39] INFO jenkins.InitReactorRunner$1$onAttained: Started all plugins
2026-07-15 08:42:04.113+0000 [id=39] INFO jenkins.InitReactorRunner$1$onAttained: Augmented all extensions
2026-07-15 08:42:04.259+0000 [id=36] INFO jenkins.InitReactorRunner$1$onAttained: System config loaded
2026-07-15 08:42:04.260+0000 [id=32] INFO jenkins.InitReactorRunner$1$onAttained: System config adapted
2026-07-15 08:42:04.261+0000 [id=40] INFO jenkins.InitReactorRunner$1$onAttained: Loaded all jobs
2026-07-15 08:42:04.263+0000 [id=40] INFO jenkins.InitReactorRunner$1$onAttained: Configuration for all jobs updated
2026-07-15 08:42:04.385+0000 [id=69] INFO hudson.util.Retrier$start: Attempt #1 to do the action check updates server
2026-07-15 08:42:04.594+0000 [id=47] INFO jenkins.install.SetupWizard$init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 8c28ae16215b4cad85e5e8f1e436c1b
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
2026-07-15 08:42:10.467+0000 [id=47] INFO jenkins.InitReactorRunner$1$onAttained: Completed initialization
2026-07-15 08:42:10.484+0000 [id=25] INFO hudson.lifecycle.Lifecycle$onReady: Jenkins is fully up and running
2026-07-15 08:42:12.051+0000 [id=69] INFO h.n.DownloadService$downloadableLoad: Obtained the updated data file for hudson tasks Maven.MavenInstaller
2026-07-15 08:42:12.051+0000 [id=69] INFO hudson.util.Retrier$start: Performed the action check updates server successfully at the attempt #1
PS C:\WINDOWS\system32>
```

My courses | SIT

Announcement: x

Devops Lab - Se x

DEVOPS LAB 1 x

Dockerizing Fla: x

Sign in - Jenkins x

WhatsApp x

Ask Gemini

localhost:8080/login?from=%2F

Getting Started

Unlock Jenkins

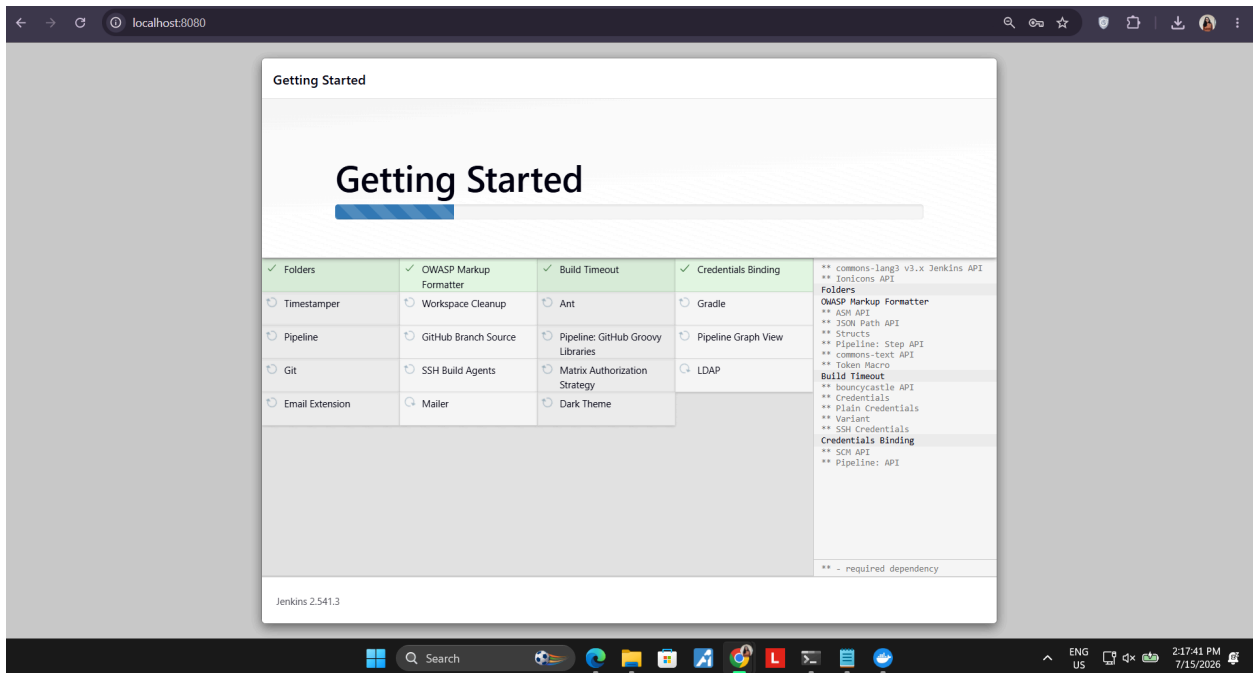
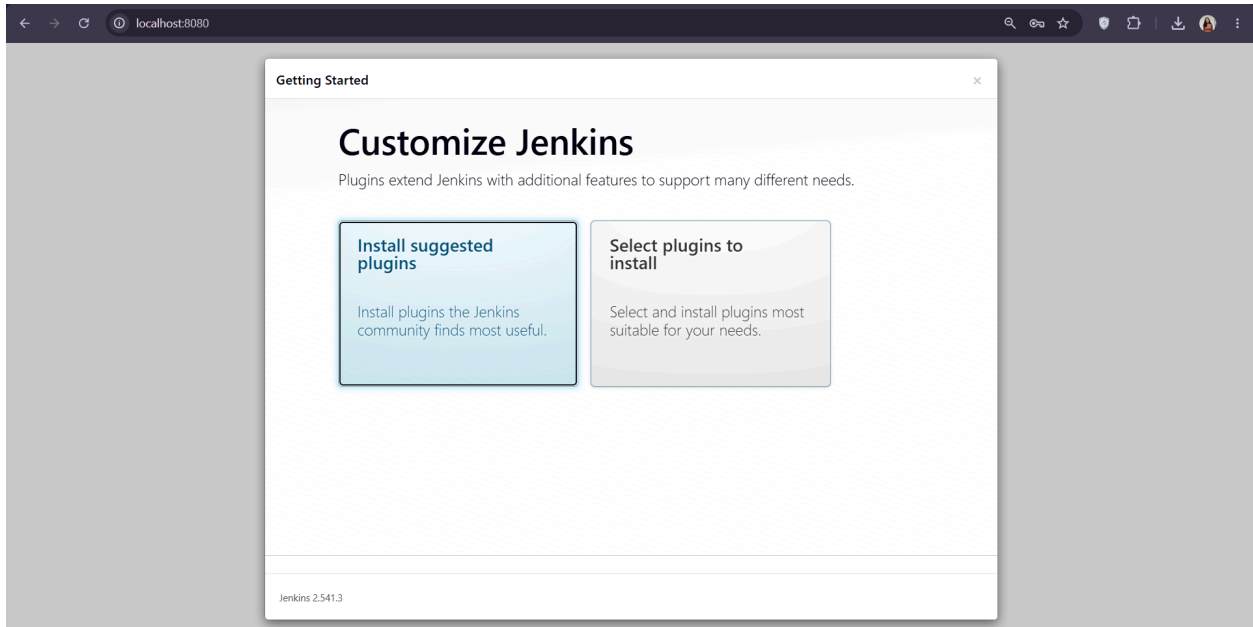
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

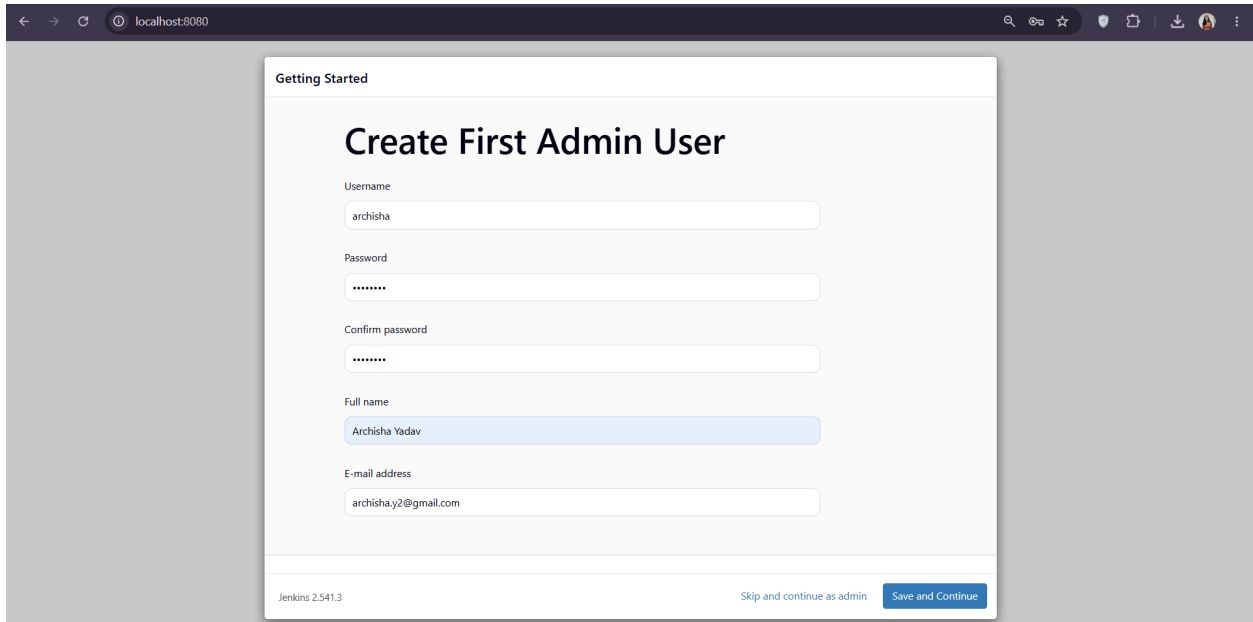
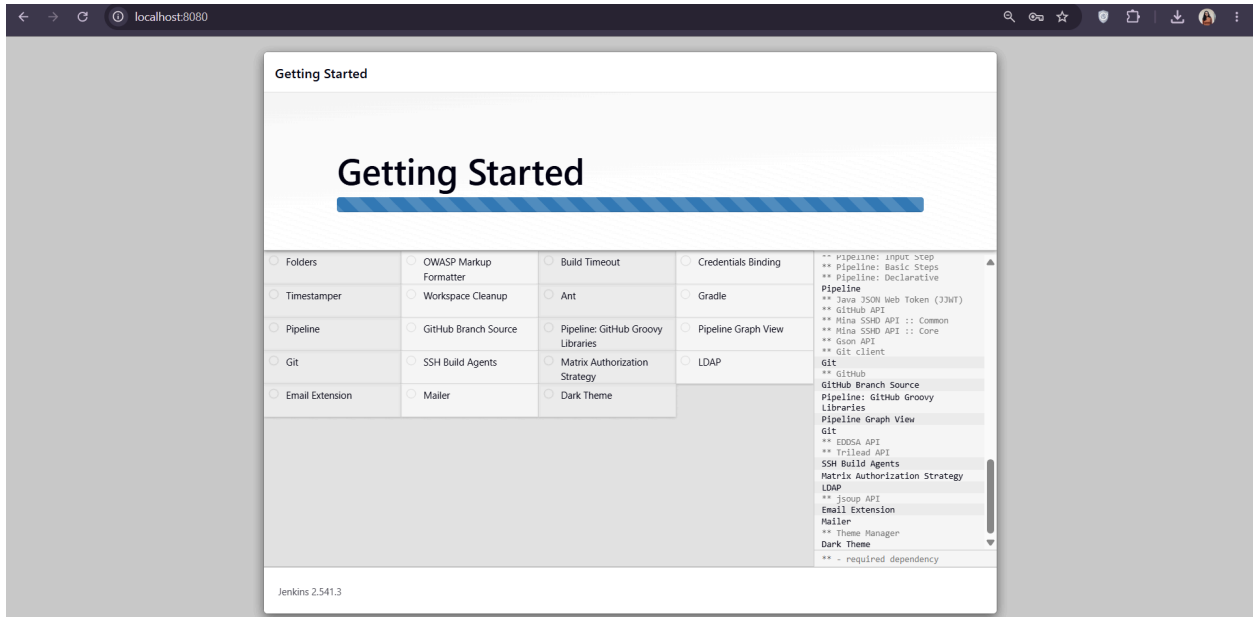
```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue





Getting Started

Instance Configuration

Jenkins URL:

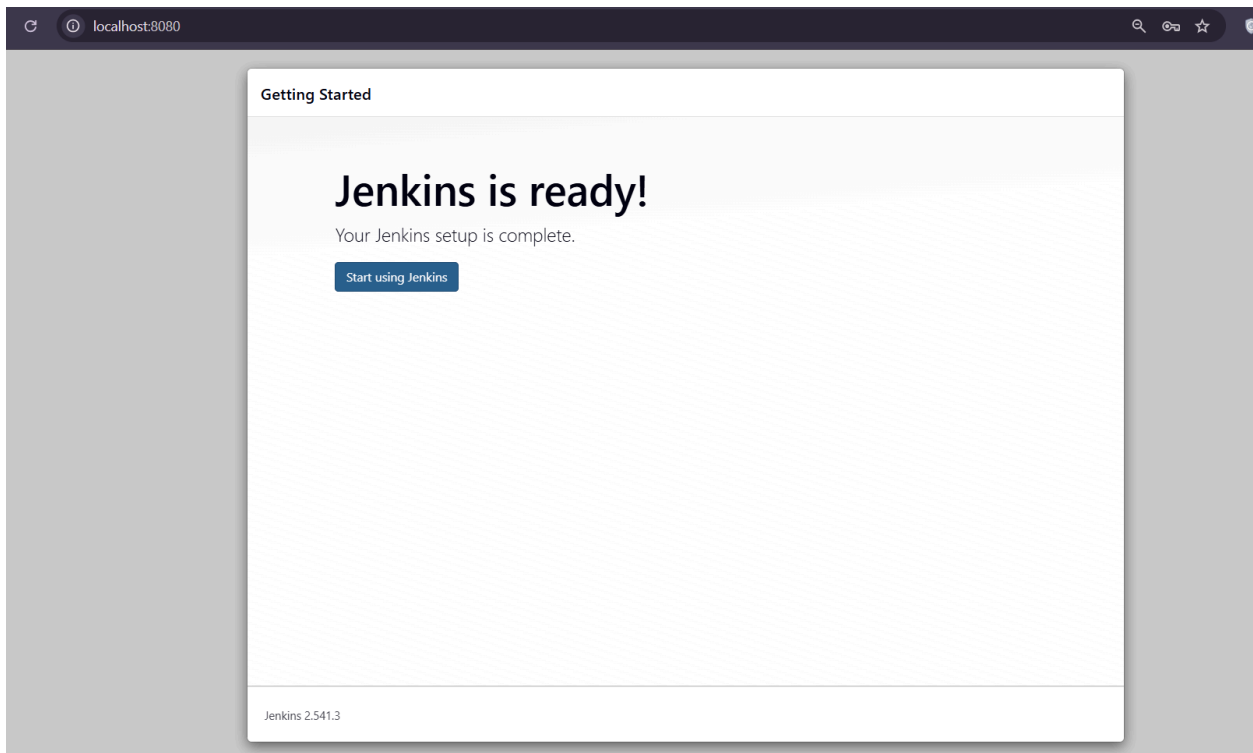
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the `BUILD_URL` environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.541.3

Not now

Save and Finish



← → ↻ 📄 localhost:8080

🔍 ☆ 🛡️ 📄 ⬇️ 👤 ⋮

 **Jenkins**

+ New Item

📅 Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

🔍 ⚙️ 👤

Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API

Jenkins 2.541.3

← → ↻ 📄 localhost:8080/manage/pluginManager/installed

🔍 ☆ 🛡️ 📄 ⬇️ 👤 ⋮

 **Jenkins**

Manage Jenkins ▾ / Plugins

📄 Updates

📅 Available plugins

📅 Installed plugins

⚙️ Advanced settings

📊 Download progress

🔍 git plugin

Name 1	Health	Enabled
Git client plugin 6.6.1 Utility plugin for Git support in Jenkins Report an issue with this plugin	100	
Git plugin 5.10.1 This plugin integrates Git with Jenkins. Report an issue with this plugin	100	
GitHub API Plugin 1.330-492.v3941a_032db_2a_ This plugin provides GitHub API for other plugins. Report an issue with this plugin	97	
GitHub Branch Source Plugin 1967.1970.vd86979736546 Multibranch projects and organization folders from GitHub. Maintained by CloudBees, Inc. Report an issue with this plugin	97	
GitHub plugin 1.47.0 This plugin integrates GitHub to Jenkins. Report an issue with this plugin	97	
Pipeline: GitHub Groovy Libraries 65.v203688e7727e Allows Pipeline Groovy libraries to be loaded on the fly from GitHub. Report an issue with this plugin	92	

REST API

Jenkins 2.541.3

New Item

Enter an item name

Git-Practice-Build

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.



Organization Folder

OK

localhost:8080/job/Git-Practice-Build/configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/adroitathena2/my-git-practice.git

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Save Apply

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

☐ Delete workspace before build starts

☐ Use secret text(s) or file(s) ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

+ Add build step

Filter

- Execute Windows batch command
- Execute shell
- Invoke Ant
- Invoke Gradle script
- Invoke top-level Maven targets
- Run with timeout
- Set build status to "pending" on GitHub commit

REST API Jenkins 2.541.3

 Jenkins

Git-Practice-Build

Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell

Command

See the list of available environment variables

```
echo "-----"
echo "BUILDING: ${JOB_NAME}, BUILD #${BUILD_NUMBER}" # Use Jenkins
environment variables
echo "Workspace: ${WORKSPACE}"
echo "-----"
echo "Listing files checked out from Git:"
ls -la
echo "-----"
echo "Contents of README.md:"
cat README.md
echo "-----"
echo "Hello Jenkins!"
```

Advanced

+ Add build step

Save Apply

 Jenkins

Git-Practice-Build

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds >

No builds

Git-Practice-Build

Permalinks

Add description

localhost:8080/job/Git-Practice-Build-New/1/

Jenkins / Git-Practice-Build-New / #1

Status

Changes

Console Output

Edit Build Information

Delete build #1

Git Build Data

#1 (Jul 15, 2026, 9:44:45 AM)

Add description

Keep this build forever

Started 31 sec ago
Took 5.2 sec

Started by user Archisha Yadav

Revision: 128f121c11b71b4a1bf26b7ff0d2bba871f5b844
Repository: <https://github.com/adroitathena2/my-git-practice.git>

- refs/remotes/origin/main

No changes.

REST API Jenkins 2.541.3

localhost:8080/job/Git-Practice-Build-New/1/console

Jenkins / Git-Practice-Build-New / #1

Status

Changes

Console Output

Edit Build Information

Delete build #1

Git Build Data

Console

Download

Copy

View as plain text

```
Started by user Archisha Yadav
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Git-Practice-Build-New
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/adroitathena2/my-git-practice.git
> git init /var/jenkins_home/workspace/Git-Practice-Build-New # timeout=10
Fetching upstream changes from https://github.com/adroitathena2/my-git-practice.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/adroitathena2/my-git-practice.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/adroitathena2/my-git-practice.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^[commit] # timeout=10
Checking out Revision 128f121c11b71b4a1bf26b7ff0d2bba871f5b844 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 128f121c11b71b4a1bf26b7ff0d2bba871f5b844 # timeout=10
Commit message: "Create helloworld.py"
First time build. Skipping changelog.
[Git-Practice-Build-New] $ /bin/sh -xe /tmp/jenkins4702508998972529842.sh
+ echo -----
+ echo BUILDING: Git-Practice-Build-New, BUILD #1
BUILDING: Git-Practice-Build-New, BUILD #1
+ echo Workspace: /var/jenkins_home/workspace/Git-Practice-Build-New
Workspace: /var/jenkins_home/workspace/Git-Practice-Build-New
```

<https://github.com/adroitathena2/my-git-practice.git>


```
localhost:8080/job/Git-Practice-Build-New/1/console

Jenkins / Git-Practice-Build-New / #1

+ echo BUILDING: Git-Practice-Build-New, BUILD #1
BUILDING: Git-Practice-Build-New, BUILD #1
+ echo Workspace: /var/jenkins_home/workspace/Git-Practice-Build-New
Workspace: /var/jenkins_home/workspace/Git-Practice-Build-New
+ echo -----
+ echo Listing files checked out from Git:
Listing files checked out from Git:
+ ls -la
total 20
drwxr-xr-x 3 jenkins jenkins 4096 Jul 15 09:44 .
drwxr-xr-x 5 jenkins jenkins 4096 Jul 15 09:44 ..
drwxr-xr-x 8 jenkins jenkins 4096 Jul 15 09:44 .git
-rw-r--r-- 1 jenkins jenkins 53 Jul 15 09:44 README.md
-rw-r--r-- 1 jenkins jenkins 21 Jul 15 09:44 helloworld.py
+ echo -----
+ echo Contents of README.md:
Contents of README.md:
+ cat README.md
# My Git Practice Project

Editing this README.md

+ echo -----
+ echo Hello Jenkins!
Hello Jenkins!
Finished: SUCCESS
```

Basic Declarative Pipeline

localhost:8080/view/all/newJob

Jenkins / All / New Item

New Item

Enter an item name

Declarative-Pipeline-Demo

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

OK



Jenkins / Declarative-Pipeline-Demo ▾ / Stages

Stages

July 16, 2026



#1

14:42 - 8s

Checkout Source Code

5s



← → ↺ ⓘ localhost:8080/job/Declarative-Pipeline-Demo/1/console

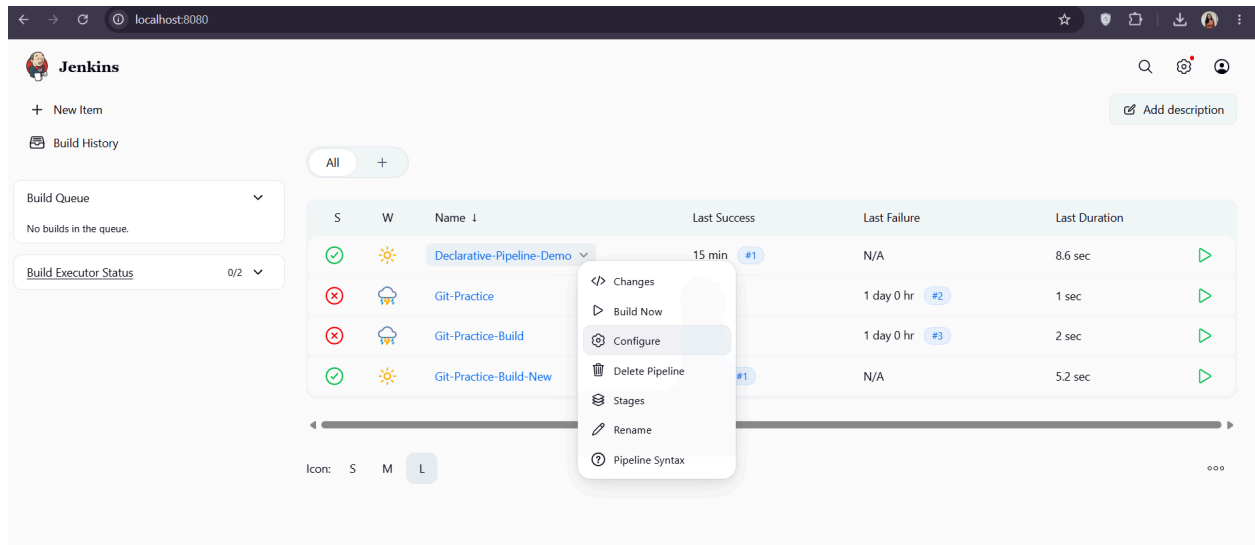


Jenkins / Declarative-Pipeline-Demo ▾ / #1 ▾

```
[Pipeline] { (Run Example Command)
[Pipeline] echo
Executing a shell command...
[Pipeline] sh
+ echo Hello from Declarative Pipeline!
Hello from Declarative Pipeline!
[Pipeline] sh
+ echo --- Reading README ---
--- Reading README ---
[Pipeline] sh
+ cat README.md
# My Git Practice Project
```

Editing this README.md

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



```

pipeline {
  agent any // 1. Define Agent: Run on any available Jenkins agent/executor

  // Add the parameters block here, outside of stages
  parameters {
    string(name: 'GREETING_NAME', defaultValue: 'World', description: 'The name to use in
the greeting')
  }

  stages { // 2. Define Stages: Container for all the work phases

    // New stage to test the parameter output
    stage('Greeting') {
      steps {
        // Double quotes are required here for Groovy string interpolation
        echo "Hello ${params.GREETING_NAME}"
      }
    }

    stage('Checkout Source Code') { // 3. Define Stage 1: Checkout
      steps { // 4. Define Steps for Stage 1
        echo 'Starting Git checkout...'
        git url: 'https://github.com/adroitathena2/my-git-practice.git', branch: 'main'

        // If your repo is private, you might need:
        // credentialsId: 'your-jenkins-credential-id'

        echo 'Git checkout completed.'
      }
    }
  }
}

```

```

stage('Run Example Command') { // 5. Define Stage 2: Run Command
  steps { // 6. Define Steps for Stage 2
    echo 'Executing a shell command...'

    // Use 'sh' for Linux/macOS agents, 'bat' for Windows
    sh 'echo "Hello from Declarative Pipeline!"'
    sh 'echo "--- Reading README ---"'
    sh 'cat README.md' // Verify checkout by reading the file
  }
}

// You could add more stages here (e.g., Build, Test, Deploy)

} // End of stages

// Optional: Add post actions that run after all stages
// post {
//   always {
//     echo 'Pipeline finished.'
//   }
// }
} // End of pipeline

```

localhost:8080/job/Declarative-Pipeline-Demo/

Jenkins / Declarative-Pipeline-Demo

Status

Changes

Build with Parameters

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

Declarative-Pipeline-Demo

Permalinks

- Last build (#2), 4 min 20 sec ago
- Last stable build (#2), 4 min 20 sec ago
- Last successful build (#2), 4 min 20 sec ago
- Last completed build (#2), 4 min 20 sec ago

Builds >

Filter

Today

- #2 9:34 AM
- #1 9:12 AM

localhost:8080/job/Declarative-Pipeline-Demo/2/console

Jenkins / Declarative-Pipeline-Demo / #2

```
> git branch -D main # timeout=10
> git checkout -b main 128f121c11b71b4a1bf26b7ff0d2bba871f5b844 # timeout=10
Commit message: "Create helloworld.py"
> git rev-list --no-walk 128f121c11b71b4a1bf26b7ff0d2bba871f5b844 # timeout=10
[Pipeline] echo
Git checkout completed.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Example Command)
[Pipeline] echo
Executing a shell command...
[Pipeline] sh
+ echo Hello from Declarative Pipeline!
Hello from Declarative Pipeline!
[Pipeline] sh
+ echo --- Reading README ---
--- Reading README ---
[Pipeline] sh
+ cat README.md
# My Git Practice Project

Editing this README.md

[Pipeline] }
[Pipeline] // stage
... ..
```

(Done till pg 29)

Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Archisha>cd Documents\semVII\Devops Lab

C:\Users\Archisha\Documents\semVII\Devops Lab>git clone
https://github.com/aditisharmas11/Devops-Lab-L1_2023-27.git
Cloning into 'Devops-Lab-L1_2023-27'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 12 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), done.

C:\Users\Archisha\Documents\semVII\Devops Lab>cd Devops-Lab-L1_2023-27

C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git checkout -b
project1_ArchishaYadav_23070122041
Switched to a new branch 'project1_ArchishaYadav_23070122041'

C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>echo "# My Git
Practice Project" > README.md

C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git status
On branch project1_ArchishaYadav_23070122041
Changes not staged for commit:
 (use "git add <file>..." to update what will be committed)
 (use "git restore <file>..." to discard changes in working directory)
 modified: README.md

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git add
README.md

C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git status
On branch project1_ArchishaYadav_23070122041
Changes to be committed:
 (use "git restore --staged <file>..." to unstage)
 modified: README.md

C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git commit -m
"Add initial README.md file"

[project1_ArchishaYadav_23070122041 fe5f5f1] Add initial README.md file
1 file changed, 1 insertion(+), 1 deletion(-)

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git push -u origin
project1_ArchishaYadav_23070122041
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'project1_ArchishaYadav_23070122041' on GitHub by visiting:
remote:
https://github.com/aditisharmas11/Devops-Lab-L1_2023-27/pull/new/project1_ArchishaYadav_2
3070122041
remote:
To https://github.com/aditisharmas11/Devops-Lab-L1_2023-27.git
* [new branch]    project1_ArchishaYadav_23070122041 ->
project1_ArchishaYadav_23070122041
branch 'project1_ArchishaYadav_23070122041' set up to track
'origin/project1_ArchishaYadav_23070122041'.
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git status
On branch project1_ArchishaYadav_23070122041
Your branch is up to date with 'origin/project1_ArchishaYadav_23070122041'.
```

nothing to commit, working tree clean

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git checkout -b
feature-update
Switched to a new branch 'feature-update'
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git branch
Devops_LabL1_2023-27
* feature-update
project1_ArchishaYadav_23070122041
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git add
README.md
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git commit -m
"Update README on feature branch"
On branch feature-update
nothing to commit, working tree clean
```



```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git status
On branch feature-update
nothing to commit, working tree clean
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git checkout
project1_ArchishaYadav_23070122041
Switched to branch 'project1_ArchishaYadav_23070122041'
Your branch is up to date with 'origin/project1_ArchishaYadav_23070122041'.
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git merge
feature-update
Already up to date.
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git push origin
project1_ArchishaYadav_23070122041
Everything up-to-date
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git add
README.md
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git commit -m
"Update README first line on assignment branch"
On branch project1_ArchishaYadav_23070122041
Your branch is up to date with 'origin/project1_ArchishaYadav_23070122041'.
```

```
nothing to commit, working tree clean
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git checkout -b
conflicting-feature HEAD~1
Switched to a new branch 'conflicting-feature'
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git add
README.md
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git commit -m
"Update README first line on feature branch"
On branch conflicting-feature
nothing to commit, working tree clean
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git checkout
project1_ArchishaYadav_23070122041
Switched to branch 'project1_ArchishaYadav_23070122041'
Your branch is up to date with 'origin/project1_ArchishaYadav_23070122041'.
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>git merge  
conflicting-feature  
Already up to date.
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27><<<<<< HEAD  
<< was unexpected at this time.
```

```
C:\Users\Archisha\Documents\semVII\Devops Lab\Devops-Lab-L1_2023-27>
```